



Instructor: Date:

Learner: Regional Branch:

Program: Center:

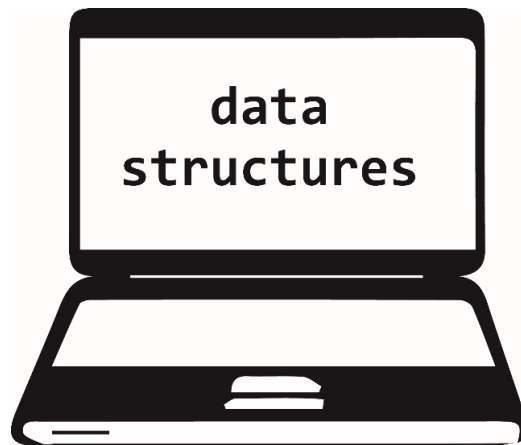
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Let's Start!

What are data structures?



Learning Outcome:

Describe common activities at work to a visitor using short sentences in familiar spoken and written situations.
Understand information about data and its structure in simple spoken and written situations.

Can Do Statements:

- I can explain how data structures work.
- I can give basic information about data structures.
- I can describe key features related to data structures.

Ask your partner:

- How important is organizing information in a database?
- How do you think data can be arranged to improve efficiency?





Let's Explore!

Francine and Laura talk about data structures in a company.



Instructions: Read the text below.

- Francine:** Hello, Laura. Are you organizing the conference for this afternoon? I heard it's going to be very interesting.
- Laura:** Yes, you must come before 2 pm to get a seat. We are covering very important topics related to database management.
- Francine:** Unfortunately, I can't go. I have to take my daughter to the dentist. I couldn't reschedule her appointment. Are you talking about data mining?
- Laura:** No, we aren't focusing on that issue today, but we're discussing data structures.
- Francine:** That's too bad! I am not going to be able to solve some doubts I have. Can you briefly tell me what data structures are?
- Laura:** Sure! As you must know, a data structure is the easiest way to organize data on your computer. Its principal components are some series of elementary data types, a system to transfer these data, and particular strategies to manipulate them.
- Francine:** Could you explain to me how data structures work?
- Laura:** They work through arrays, which don't have internal links that join their elements because all data have to belong to the same category. On the other hand, they also work through types of lists, which have more than one pointer to join elements. These types of lists can have different functions, such as tree lists. A tree list starts with a root element or an element that doesn't have a predecessor. In this way, the next elements will only have one predecessor. This is an advantage since the process to execute systematic operations becomes faster.
- Francine:** That's an interesting explanation! I'm not going to forget it. What else are you going to discuss about data structures?
- Laura:** Well, we have to discuss many important aspects; however, I am not going to mention all of them, only the most common key features. There's another type of list called queue, this one retrieves data in a "first in, first out" structure. For example, when there are tasks that have to wait their turn for execution. Finally, we find the stack lists, in this case, you can only access them at the top, so when you need to insert a new item, you're going to remove the last inserted item. In other words, this is a "last in, first out" logic when organizing and retrieving elements.

**Let's Practice!**

Laura explains what data structures are.

Instructions: Select the correct option to correctly complete each sentence.

1. You must (to know / knowing / know) that a data structure works through different types of lists, such as arrays, queues, and stacks.
2. We (can to / can / could to) create a tree list to organize better those elements.
3. When working with stack lists, you have (to access / access / accessed) them at the top since each new item is going to remove the last inserted item.

**Let's Practice!**

What else can you tell me about data structures?

Instructions: Circle the correct option to complete the sentences. Read the sentences aloud. Check your pronunciation with your partner.

1. Are (coming to / you coming to) the conference about data structures at 2 pm?
2. There are many data structures. However, I'm not (mentioning / going to mention) all of them. For example, array structure is used when we know the number of elements in a database.
3. Are (you going to / you going) use the stack structure to organize and retrieve elements?



Extension Activity

Let's learn about other data structures!

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Search for more information about other data structures.
- Identify advantages and disadvantages of each data structure.
- Make a comparative chart explaining advantages and disadvantages of each data structure.
- Use the grammar and vocabulary you studied in this unit.
- Share your comparative chart in class and discuss about the topic.

Self-Assessment

- I search for more information about other data structures.
- I identify advantages and disadvantages of each data structure.
- I make a comparative chart explaining advantages and disadvantages of each data structure.
- I use the grammar and vocabulary I learned.
- I share information about data structures and discuss it with my class.

Yes

No

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Target Words

- array** (noun): order of elements in particular groups
e.g. They work through arrays, which don't have internal links that join their elements because all data have to belong to the same category.
- pointer** (noun): something that allows the quick identification of an element in a database
e.g. They also work through types of lists, which have more than one pointer to join elements.
- briefly** (adverb): in short
e.g. Can you briefly tell me what data structures are?
- reschedule** (verb): to change the time of an appointment or date
e.g. I couldn't reschedule her appointment.
- issue** (noun): a matter to discuss
e.g. We aren't focusing on that issue today, but we're discussing about data structures.
- retrieve** (verb): to search for an element in a database and then display it on the screen to check it
e.g. There's another type of list called queue, which retrieve data in a "first in, first out" structure.



Learning Points

Present Continuous vs. Be Going to in Questions and Negative Sentences

Apart from the simple future with "will," we can also use the **present continuous tense** and the future with be going to in order to talk about future activities.

You use both the present continuous tense and "be going to" to express future actions. However, there are certain differences between the two forms.

We use the **present continuous tense** when we describe our plans or arrangements for a specific time in the future. These are generally fixed plans with a determined time and place. When making negative sentences and questions, the present continuous for future arrangements has the following structures:

Subject + **auxiliary verb to be** + **not** + **main verb** + **ing** + complement

Example:

- *That data structure **is not using** a tree list.*

Auxiliary verb to be + subject + **main verb** + **ing** + Complement + ?

Example:

- ***Is** that data structure **using** a tree list?*
Yes, it is.
No, it isn't.

On the other hand, we use "**be going to**" when we want to express our purpose or decision to do something. However, it does not imply fixed plans or arrangements, it is simply an intention. When making negative sentences and questions, "be going to" has the following structures:

Subject + **auxiliary verb to be** + **not** + **going to** + **main verb** + complement

Example:

- *Ronald **isn't going to** validate data structures today.*

Auxiliary verb to be + subject + **going to** + **main verb** + complement + ?





Learning Points

Example:

- ***Is** Ronald **going to** validate data structures today?*
Yes, he **is**.
No, he **isn't**.

Examples

- *Are you organizing the conference for this afternoon?*
- *No, we aren't focusing on that issue today, but we're discussing data structures.*
- *That's too bad! I am not going to be able to solve some doubts I have.*
- *What else are you going to discuss about data structures?*

Modal Verbs Can, Could, Have to and Must

We use the modal verbs **can** and **could** to express abilities and possibilities. When we want to express obligation we use the modals **have to** and **must**.

The modals **can** and **could** are useful to express skills and possible actions. We use **can** in the present and **could** in the past.

Structure:

Subject + **can/could** + **main verb in base form** + complement

Examples:

- *You **can try** many data structures and then apply the one that fits better with your purpose.*
- *We **could organize** those items in a tree line structure.*

The modal verbs **have to** and **must** are used when referring to obligations. However, there are certain differences to consider.

We use **must** to express a strong obligation that is related to a personal belief or circumstance. It implies certain agreement from the speaker.





Learning Points

Structure:

Subject + **must** + **main verb in base form** + complement

Example:

- We **must know** first each type of data structure before making a decision.

We use **have to** in order to talk about a strong obligation that is related to external circumstances. The speaker doesn't necessarily agree with the obligation.

Structure:

Subject + **have to/has to** + **main verb in base form** + complement

Examples:

- In a tree structure, each list **has to operate** through a root element.
- We **have to organize** the company's data in an easier manner.

Examples

- Can you briefly tell me what data structures are?
- I couldn't reschedule her appointment.
- As you must know, a data structure is the easiest way to organize data on your computer.
- For example, when there are tasks that have to wait their turn for execution.